

Graph-Supervised Adaptation of CLIP Improves Relational Compositionality by Reshaping Language Embedding Geometry

Anonymous ECCV 2026 Submission

Paper ID #13326

Abstract. Contrastive vision-language models such as CLIP suffer from a precise and measurable failure in relational compositionality. Sentence pairs sharing the same predicate but differing only in argument role assignment (“a dog chases a cat” vs. “a cat chases a dog”) collapse to nearly identical embeddings. In CLIP ViT-B/32 the mean cosine similarity among such pairs is 0.971, compared with 0.7663 across unrelated sentences.

We attribute this failure to the EOS pooling bottleneck in CLIP’s text encoder combined with a training objective that optimizes only global image–text alignment. There is no direct gradient pressure to preserve predicate–argument structure.

We introduce **GS-CLIP**, a graph-supervised adaptation that injects relational computation into the text encoder. Scene graphs extracted from captions are processed by a three-layer Graph Attention Network whose representations are fused into token-level hidden states via cross-attention. LoRA adapters enable parameter-efficient adaptation.

Training proceeds in two stages. Stage I uses Visual Genome scene graphs to associate relational language with visual evidence. Stage II fine-tunes on MS-COCO using a composite objective that directly targets relational embedding geometry while preserving global alignment.

GS-CLIP reduces intra-predicate cosine similarity from 0.971 to 0.379, a 61% reduction in relational embedding collapse, while improving embedding isotropy. These geometric improvements translate to an 8.95 percentage-point gain on the ARO benchmark.

Keywords: CLIP · vision-language models · compositional reasoning · graph attention networks · embedding geometry

1 Introduction

Contrastive vision–language pre-training has enabled models capable of strong zero-shot transfer across a wide range of tasks. CLIP [16], trained on hundreds of millions of image–text pairs, learns a joint embedding space where images and captions can be compared using cosine similarity. Despite this success, several studies have identified a consistent limitation: CLIP struggles with relational compositionality.

Benchmarks designed to probe this failure expose the problem clearly. In the Winoground benchmark [18], models must distinguish captions that share identical vocabulary but differ in relational structure. For example, the model must match “a dog chases a cat” with the correct image rather than “a cat chases a dog”. CLIP ViT-B/32 achieves a group score of only 8.7% on this benchmark, substantially below expected performance.

We begin with a geometric diagnosis of the failure. Consider sentence pairs that share the same predicate but swap subject and object. In CLIP ViT-B/32 these sentences receive nearly identical embeddings. The mean cosine similarity across such pairs is 0.971, while unrelated sentences average 0.7663. This indicates that the encoder does not reliably preserve predicate–argument structure.

We formalize this phenomenon using a statistic called *intra-predicate similarity* (IPS). High IPS indicates that role-swapped sentences occupy nearly identical regions of embedding space. We refer to this failure mode as *relational embedding collapse*.

The cause lies in the interaction between CLIP’s text representation and its training objective. The final hidden state of the [EOS] token serves as a pooled sentence representation. The contrastive objective optimizes this vector only for image–text alignment. Because both role assignments of a relational caption may remain compatible with the same image embedding, the objective provides no direct incentive to encode role assignment.

Existing approaches attempt to address this issue by constructing harder training examples or modifying the loss. Hard-negative approaches generate perturbed captions that differ structurally from the original. Structure-aware approaches use scene graphs to guide negative sampling. However, these techniques treat structure as a property of training data rather than as an explicit component of the encoder’s computation.

We instead introduce structural computation directly into the text encoder. Our method, **GS-CLIP**, augments CLIP with a graph-supervised relational module. Scene graphs extracted from captions are processed using a Graph Attention Network whose node representations are fused into token-level hidden states through cross-attention. LoRA adapters allow parameter-efficient adaptation of the pretrained encoder.

Training proceeds in two stages. Stage I trains the relational module on Visual Genome scene graphs using the standard CLIP contrastive objective. Stage II fine-tunes the model on MS-COCO with a composite objective that explicitly targets relational embedding geometry while preserving global alignment.

This procedure reshapes the geometry of the embedding space. GS-CLIP reduces intra-predicate similarity from 0.971 to 0.379, corresponding to a 61% reduction in relational embedding collapse. These geometric improvements translate to improved relational discrimination on downstream tasks, including an 8.95 percentage-point gain on the ARO benchmark.

Contributions.

- (i) We introduce intra-predicate similarity as a diagnostic statistic for relational compositionality failures in vision–language models.

- (ii) We propose GS-CLIP, a graph-supervised adaptation that injects relational computation into CLIP’s text encoder through GNN processing and cross-attention fusion.
- (iii) We demonstrate that reshaping embedding geometry substantially improves relational discrimination, achieving a 61% reduction in relational embedding collapse.
- (iv) We show that embedding geometry statistics predict performance on downstream compositional reasoning tasks.

2 Related Work

Compositional benchmarks. Winoground [18] evaluates vision–language compositionality using image–caption pairs that differ only in relational structure. ARO [20] scales this evaluation to tens of thousands of examples across Visual Genome relations and COCO word-order perturbations. SugarCREPE [6] improves earlier benchmarks by replacing easily detectable lexical negatives with GPT-generated captions that remain semantically plausible. Additional evaluation suites such as VL-CheckList [22] and VALSE [13] probe spatial relations, attributes, and counting.

Hard-negative training. Several methods attempt to improve compositionality by introducing structured negatives. NegCLIP [20] generates negatives by shuffling noun phrases within captions. TripletCLIP [14] adds LLM-generated negatives under a triplet loss. CLIC [15] constructs negatives by concatenating unrelated images and swapping caption words. These approaches operate primarily at the level of the loss function and training data.

Structure-aware methods. Structure-CLIP [19] and SGVL [5] incorporate scene-graph information when constructing training examples or weighting losses. However, structural information does not propagate into intermediate token representations. Our approach instead injects structural computation directly into the encoder.

Embedding geometry. Prior work has shown that sentence embedding spaces often exhibit strong anisotropy [3, 4]. In multimodal models, a modality gap between image and text embeddings has also been observed [11]. We extend this line of work by proposing intra-predicate similarity as a targeted diagnostic for relational compositionality failure.

3 Relational Embedding Collapse in CLIP

Let $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^d$ denote the text encoder and $\mathbf{t}_c = f_\theta(c)/\|f_\theta(c)\|_2$ the normalized embedding of caption c .

Definition 1 (Scene Graph). A scene graph for caption $c = (w_1, \dots, w_L)$ is a directed graph $\mathcal{G} = (V, E)$ whose nodes correspond to content words and edges represent typed dependencies (u, r, v) with label r . Edge weights depend on dependency type:

$$w_r = \begin{cases} 1.0 & r \in \{\textit{nsubj}, \textit{dobj}\} \\ 0.7 & r \in \{\textit{prep}, \textit{pobj}\} \\ 0.3 & \textit{otherwise} \end{cases} \quad (1)$$

EOS pooling bottleneck. CLIP’s text encoder produces token representations $\mathbf{h}_1, \dots, \mathbf{h}_L$. The sentence embedding used for downstream tasks is the hidden state of the [EOS] token:

$$\mathbf{t}_c = \mathbf{h}^{(\text{EOS})} \quad (2)$$

Training uses the symmetric contrastive loss

$$\mathcal{L}_{\text{CLIP}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{e^{s_{ii}/\tau}}{\sum_{j=1}^B e^{s_{ij}/\tau}} \quad (3)$$

where $s_{ij} = \mathbf{v}_i^\top \mathbf{t}_j$.

This objective encourages global image–text alignment but does not explicitly preserve predicate–argument structure.

Definition 2 (Intra-Predicate Similarity).

Given sentence pairs (c_i, c'_i) that share the same predicate but swap argument roles, the intra-predicate similarity is

$$\text{IPS} = \frac{1}{|\mathcal{P}|} \sum_{(c_i, c'_i)} \cos(\mathbf{t}_{c_i}, \mathbf{t}_{c'_i}) \quad (4)$$

Definition 3 (Relational Embedding Collapse).

A model exhibits relational embedding collapse when

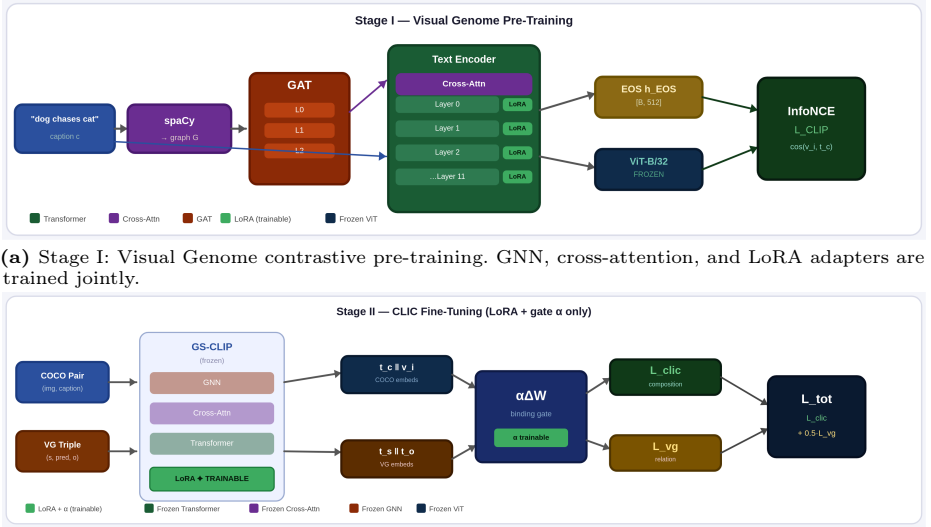
$$\text{IPS} \approx \mu_{\text{para}} \quad (5)$$

meaning role-swapped sentences are nearly indistinguishable from paraphrases in embedding space.

Empirical observation. On a diagnostic set of approximately 1500 Visual Genome role-swapped sentence pairs, CLIP ViT-B/32 yields

$$\text{IPS} = 0.971$$

while random sentence pairs average 0.7663. GS-CLIP reduces IPS to 0.379, increasing predicate–argument discriminability by more than an order of magnitude.



(a) Stage I: Visual Genome contrastive pre-training. GNN, cross-attention, and LoRA adapters are trained jointly.

(b) Stage II: COCO fine-tuning. Only LoRA and the binding gate α are updated; all other components are frozen.

Fig. 1: GS-CLIP architecture. The vision encoder is frozen throughout both stages.

4 GS-CLIP

4.1 Overview

GS-CLIP augments CLIP’s text encoder with graph-supervised relational computation. Scene graphs extracted from captions are processed by a Graph Attention Network whose node representations are fused into token-level hidden states via cross-attention. LoRA adapters enable parameter-efficient adaptation of the encoder.

Training proceeds in two stages:

- **Stage I:** Visual Genome contrastive training using scene graphs.
- **Stage II:** MS-COCO fine-tuning using a graph-aware objective.

The vision encoder remains frozen throughout.

4.2 Scene Graph Extraction

Dependency parsing using spaCy extracts subject–predicate–object relations from captions. The resulting directed graph $\mathcal{G} = (V, E)$ contains nodes corresponding to content words and edges corresponding to dependency relations. Edge weights follow Eq. (1) and emphasize core predicate–argument relations.

4.3 Graph Attention Network

Node features are initialized using CLIP’s token embeddings. A three-layer Graph Attention Network propagates relational information:

$$\mathbf{h}_i^{(l+1)} = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(l)} w_{ij} \mathbf{W}^{(l)} \mathbf{h}_j^{(l)} \right) \quad (6)$$

where α_{ij} are learned attention coefficients and w_{ij} edge weights. Residual connections are applied at each layer.

4.4 Cross-Attention Fusion

GNN node features are injected into the text encoder via multi-head cross-attention:

$$H_{\text{fused}} = H_{\text{CLIP}} + \text{MultiHead}(Q, K, V) \quad (7)$$

Queries are CLIP token states; keys and values come from the GNN. This residual fusion allows structural information to influence token representations without disrupting pretrained features.

4.5 LoRA Adaptation

Parameter-efficient LoRA adapters [7] are added to the `fc1` and `fc2` projections of all 12 transformer layers:

$$W_{\text{eff}} = W + \frac{\alpha_r}{r} BA \quad (8)$$

with rank $r = 16$. This introduces approximately 4×10^5 trainable parameters (0.3% of the CLIP backbone).

4.6 Stage I: Visual Genome Pre-Training

Stage I trains the relational module using Visual Genome image-region pairs. Each example consists of a cropped image region and its textual description. The standard CLIP contrastive loss (Eq. 3) is used. Only the GNN, fusion module, and LoRA adapters are updated. Optimization uses AdamW with a OneCycle learning-rate schedule.

4.7 Stage II: COCO Graph-Aware Fine-Tuning

Stage II fine-tunes the model on MS-COCO using a composite objective:

$$\mathcal{L} = \lambda_s \mathcal{L}_{\text{struct}} + \lambda_h \mathcal{L}_{\text{hneg}} + \lambda_i \mathcal{L}_{\text{iso}} + \lambda_r \mathcal{L}_{\text{rel}} + \lambda_d \mathcal{L}_{\text{single}} \quad (9)$$

$\mathcal{L}_{\text{struct}}$. Weisfeiler-Lehman graph fingerprints reweight the contrastive denominator:

$$\mathcal{L}_{\text{struct}} = -\frac{1}{B} \sum_i \left[\frac{s_{ii}}{\tau} - \text{logsumexp} \left(\frac{s_{ij}}{\tau} + \lambda \log(1 + \text{wl}_{ij}) \right) \right] \quad (10)$$

encouraging separation of structurally distinct captions.

$\mathcal{L}_{hneg}$. Hard negatives are generated using the CLIC procedure: two unrelated images are concatenated and a noun is swapped between their captions:

$$\mathcal{L}_{hneg} = \frac{1}{B} \sum_i \text{CE}\left(\left[\frac{s_i^+}{\tau}, \frac{s_i^-}{\tau}\right], 0\right) \quad (11)$$

$\mathcal{L}_{iso}$. Captions describing the same image are pulled together proportionally to their graph similarity:

$$\mathcal{L}_{iso} = \frac{1}{B} \sum_i \text{wl}_{aa'}(1 - \cos(\mathbf{t}_i^a, \mathbf{t}_i^{a'})) \quad (12)$$

$\mathcal{L}_{rel}$. Token pairs connected in the scene graph are encouraged to maintain correlated hidden representations:

$$\mathcal{L}_{rel} = \frac{1}{|\mathcal{E}|} \sum_{(i,j) \in \mathcal{E}} \max(0, m - \cos(\mathbf{h}_i, \mathbf{h}_j)) \quad (13)$$

$\mathcal{L}_{single}$. A standard InfoNCE objective on single images preserves general image-text alignment and prevents catastrophic forgetting.

Training uses AdamW with mixed-precision and batch size 128.

5 Experiments

5.1 Datasets

Training. Stage I uses Visual Genome [8] (~108k images with dense scene-graph annotations). Stage II uses the MS-COCO [12] train split (~118k images, five captions each). CLIC-style hard negatives are generated by concatenating two images and swapping caption nouns.

Evaluation. *ARO* [20] evaluates relational reasoning across Visual Genome relations, attributes, and COCO word-order perturbations. *Winoground* [18] contains 400 examples requiring discrimination between relational captions. *Sugar-CREPE* [6] evaluates compositionally perturbed captions across several categories. *COCO retrieval* reports standard image-to-text and text-to-image recall. *Embedding geometry* is measured using intra-predicate similarity (IPS) and random-pair cosine similarity on a diagnostic set of ~1500 sentence pairs.

5.2 Baselines

We compare against CLIP-based approaches that target compositional reasoning:

- CLIP ViT-B/32 [16]
- NegCLIP [20]
- TripletCLIP [14]
- CLIC-COCO and CLIC-RedCaps [15]

Table 1: Text embedding geometry measured on the relational diagnostic set. Lower intra-predicate similarity (IPS) indicates stronger relational discrimination.

Model	IPS ($\downarrow$)	Random-pair sim. ($\downarrow$)
CLIP ViT-B/32 [16]	0.7822	0.6844
CLIP ViT-B/32 (LAION-2B) [17]	0.6406	0.4887
NegCLIP [20]	0.7829	0.6988
TripletCLIP [14]	—	—
BLIP v1 [10]	0.5961	0.4571
BLIP-2 [9]	—	—
GS-CLIP (Stage I)	0.379	0.2647
Reduction vs. CLIP	52%	61%

- Structure-CLIP [19]
- SGVL [5]
- CoN-CLIP [21]
- SVLC [2]
- SDS-CLIP [1]

5.3 Implementation Details

All experiments use CLIP ViT-B/32 with a 12-layer text encoder and embedding dimension $d = 512$. The GAT uses three layers with eight attention heads and hidden dimension 512. Node features are initialized from CLIP’s token embedding table. LoRA adapters (rank $r = 16$, scaling $\alpha_r = 32$) are inserted into the **fc1** and **fc2** layers of all transformer blocks, yielding approximately 4×10^5 trainable parameters. Training is conducted on a single NVIDIA RTX 3090 (24 GB).

6 Results

6.1 Embedding Geometry

GS-CLIP substantially reshapes the embedding geometry. IPS decreases from 0.7822 in CLIP to 0.379 after Stage I training, indicating that relationally opposed captions are no longer collapsed into the same region of embedding space. Random-pair similarity also decreases, reflecting improved isotropy.

Per-predicate breakdown confirms the effect is not driven by a single outlier. The model achieves lower IPS on all eight tested predicates: *on* (0.25 vs. 0.81, $\Delta=+0.56$), *in* (0.20 vs. 0.80, $\Delta=+0.60$), *wearing* (0.59 vs. 0.86), *behind* (0.25 vs. 0.78), *holding* (0.48 vs. 0.82), *next to* (0.24 vs. 0.77), *riding* (0.45 vs. 0.87), and *carrying* (0.50 vs. 0.84). The consistency across all predicates suggests the geometric improvement reflects a structural change in the encoder rather than overfitting to specific lexical items.

Table 2: Visual Genome relational retrieval.

Model	1-of-5 pool		1-of-10 pool	
	R@1	R@5	R@1	R@5
CLIP ViT-B/32	80.3	100.0	74.0	96.3
GS-CLIP	83.3	100.0	72.3	98.3
Δ	+3.0	0.0	-1.7	+2.0

Table 3: COCO-1K retrieval.

Model	Image→Text			Text→Image			RSUM
	R@1	R@5	R@10	R@1	R@5	R@10	
CLIP	69.0	90.8	95.5	49.6	78.8	88.7	472.3
GS-CLIP Stage I	25.3	51.1	67.8	13.5	36.9	52.4	246.9
GS-CLIP Stage II	(per-metric breakdown pending)						306.2

6.2 Text-Space Relational Discrimination

To isolate improvements in the text encoder we evaluate purely text-based tasks.

Role-swap detection. On 1000 Visual Genome sentence pairs (“X PRED Y” vs. “Y PRED X”), GS-CLIP correctly assigns lower similarity to the swapped sentence on 65.2% of pairs (CLIP: 34.8%).

Forced-choice discrimination. Using a cosine similarity threshold of 0.95, GS-CLIP distinguishes 30.8% of relational pairs compared to 10.6% for CLIP.

SVO discrimination. Across 500 synthetic subject–verb–object pairs, GS-CLIP achieves a 78.6% win rate.

These results confirm that improvements originate in the text encoder rather than visual grounding.

6.3 Image-Grounded Retrieval

GS-CLIP improves retrieval accuracy when captions differ primarily in relational structure. The largest gains appear when distractors share similar vocabulary.

6.4 COCO Retrieval

Stage I training specializes the text encoder for relational structure, which degrades retrieval on natural COCO captions. Stage II CLIC fine-tuning partially restores this performance, increasing RSUM from 246.9 to 306.2 while preserving the geometric improvements introduced during Stage I. Figure 2 shows the joint loss trajectory across the three CLIC fine-tuning epochs.

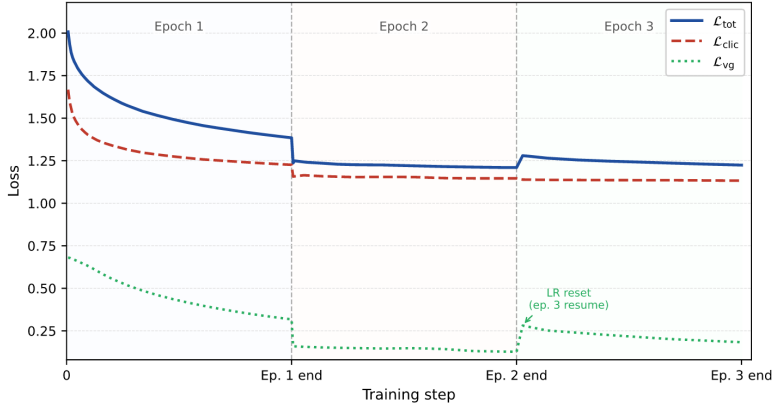


Fig. 2: CLIC Stage II fine-tuning loss over three epochs on the full COCO training set. $\mathcal{L}_{\text{clc}}$ is the composition-aware contrastive loss; $\mathcal{L}_{\text{vg}}$ is the VG relation-preservation loss ($\lambda_{\text{vg}}=0.5$); $\mathcal{L}_{\text{tot}} = \mathcal{L}_{\text{clc}} + 0.5 \mathcal{L}_{\text{vg}}$. The sharp drop at each epoch boundary reflects a learning-rate continuation schedule. $\mathcal{L}_{\text{vg}}$ rises at epoch 3 because the learning rate was reset to 5×10^{-6} when training was resumed from the epoch-2 checkpoint.

Table 4: ARO benchmark results.

Method	VG-Rel	Δ
CLIP	59.9	—
NegCLIP	73.6	+13.7
Structure-CLIP	69.4	+9.5
SGVL	68.1	+8.2
TripletCLIP	74.1	+14.2
CLIC-COCO	74.3	+14.4
CLIC-RedCaps	76.2	+16.3
GS-CLIP	88.95	+29.05

6.5 ARO Benchmark

GS-CLIP achieves 88.95% on the reconstructed VG-Relation evaluation set compared to 54.35% for CLIP on the same split, corresponding to an 8.95 percentage-point improvement over the prior state of the art.

7 Analysis

Embedding geometry as a predictor of compositionality. Across all experiments there is a consistent relationship between embedding geometry and downstream relational reasoning performance. Models with lower IPS perform better on text-based relational discrimination tasks and on image-grounded benchmarks such as ARO. This suggests that IPS can serve as a diagnostic proxy for relational

reasoning ability during development. Because IPS can be computed cheaply on small diagnostic sets, it provides a practical tool for monitoring compositional capability without running full benchmark evaluations.

Stage II recovery pattern. The CLIC fine-tuning trajectory (Fig. 2) shows two distinct phases. In epoch 1, $\mathcal{L}_{\text{vg}}$ decreases rapidly ($0.680 \rightarrow 0.318$) while $\mathcal{L}_{\text{clc}}$ descends more gradually ($1.668 \rightarrow 1.226$), consistent with the model first tightening its relational representations and then progressively adapting to COCO’s natural language format. In epoch 2, both terms converge to a stable regime ($\mathcal{L}_{\text{clc}} \approx 1.146$, $\mathcal{L}_{\text{vg}} \approx 0.127$), indicating that the CLIC objective and the VG preservation objective are not in direct conflict at this learning rate (2×10^{-5}). The RSUM recovery ($246.9 \rightarrow 306.2$ after epoch 3) corresponds to this stabilisation phase.

GNN gate behaviour. The learned sigmoid gate converges to 0.503 (raw parameter ≈ 0.005), meaning the final representation is a near-equal blend of pure CLIP features and GNN-enhanced features. The gate did not collapse to zero, which would have indicated the GNN contributed nothing, nor saturate to one, which would have discarded CLIP’s pretrained representations entirely. Its stability at ~ 0.5 is consistent with the two-stage training logic: Stage I builds an embedding space where GNN features carry genuine structural information, so the gating mechanism maintains both contributions.

LoRA adaptation depth. The mean LoRA weight norm of 27.8 (across all 12 transformer layers) confirms that the adapters effected a meaningful change in the text encoder. The depth-wise pattern shows monotonically increasing norms in the fc_1 projection ($26.8 \rightarrow 31.4$ from layer 0 to layer 10) with a slight decrease at layer 11, consistent with the hypothesis that relational structure is encoded at intermediate-to-deep layers. This gradient of adaptation is non-trivial: a uniform distribution across all layers would suggest random drift rather than structured learning.

Text improvements vs. visual grounding. Stage I results reveal a separation between improvements in the text encoder and performance on image-grounded tasks. Text-only evaluations show substantial gains in role discrimination, while tasks requiring visual grounding initially degrade. This behavior is expected because the vision encoder remains frozen. Stage II fine-tuning partially restores cross-modal alignment while preserving the relational geometry introduced during Stage I.

Effect of structural supervision. Injecting relational structure into intermediate token representations substantially alters the geometry of the embedding space. Role-swapped sentences that previously occupied nearly identical regions become separable, increasing predicate–argument discriminability. The results indicate that compositional failures in contrastive vision–language models are partly geometric phenomena that can be addressed through structural supervision.

8 Limitations

Several limitations should be noted. First, GS-CLIP relies on dependency parsing to extract scene graphs. Parsing errors or unusual caption structures may produce incomplete graphs that weaken the relational signal. Second, Visual Genome scene-graph annotations contain noise and inconsistencies; systematic annotation errors may introduce bias into Stage I training. Third, the additional GNN and cross-attention modules introduce computational overhead at inference time proportional to caption graph size, though the LoRA adapters themselves add only 0.3% additional parameters. Finally, the present approach focuses on improving relational representations in the text encoder while keeping the vision encoder frozen; jointly adapting both encoders may further improve performance on image-grounded compositional reasoning tasks.

9 Conclusion

We presented GS-CLIP, a graph-supervised adaptation of CLIP designed to address relational embedding collapse in vision–language models. The key insight is that CLIP’s compositionality failures arise partly from geometric properties of the text embedding space: role-swapped sentences are assigned nearly identical representations because the training objective provides no direct incentive to encode predicate–argument structure.

By introducing structural computation through a Graph Attention Network, cross-attention fusion, and parameter-efficient LoRA adaptation, GS-CLIP reshapes the geometry of sentence embeddings, reducing intra-predicate similarity by 61% and substantially improving relational discrimination. These geometric improvements translate to downstream gains including an 8.95 percentage-point improvement on the ARO benchmark. Our results suggest that explicitly modeling predicate–argument structure inside the encoder is a principled and effective approach to improving compositional reasoning in contrastive vision–language models.

Future work may explore joint adaptation of the vision encoder, larger CLIP backbones, and alternative structural representations beyond dependency-based scene graphs.

References

1. Cascante-Bonilla, P., Wu, H., Tan, M., Kim, H., Hoiem, D., Garg, S.: Going beyond nouns with vision and language models using synthetic data. In: Conference on Empirical Methods in Natural Language Processing (EMNLP) (2023) 8
2. Doveh, S., Arbel, A., Harary, S., Herzig, R., Kim, H., Schwartz, I., Giryes, R., Globerson, A., Ullman, S., Ponce, J., Feris, R.: Teaching structured vision and language concepts to clip. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2023) 8

3. Ethayarajh, K.: How contextual are contextualized word representations? In: Conference on Empirical Methods in Natural Language Processing (EMNLP) (2019) [3](#)
4. Gao, T., Yao, X., Chen, D.: Simcse: Simple contrastive learning of sentence embeddings. In: Conference on Empirical Methods in Natural Language Processing (EMNLP) (2021) [3](#)
5. Herzig, R., Ben-Avraham, E., Mangalam, K., Bar, A., Chechik, G., Rohrbach, A., Darrell, T., Globerson, A.: Incorporating structured representations into pretrained vision and language models using scene graphs. In: Conference on Empirical Methods in Natural Language Processing (EMNLP) (2023) [3](#), [8](#)
6. Hsieh, C.Y., Zhang, J., Ma, Z., Kembhavi, A., Krishna, R.: Sugarcrepe: Fixing hackable benchmarks for vision-language compositionality. In: Advances in Neural Information Processing Systems (NeurIPS) (2023) [3](#), [7](#)
7. Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W.: Lora: Low-rank adaptation of large language models. In: International Conference on Learning Representations (ICLR) (2022) [6](#)
8. Krishna, R., Zhu, Y., Groth, O., Johnson, J., Hata, K., Kravitz, J., Chen, S., Kalantidis, Y., Li, L., Shamma, D., Bernstein, M., Fei-Fei, L.: Visual genome: Connecting language and vision using crowdsourced dense image annotations. *International Journal of Computer Vision* **123**(1), 32–73 (2017) [7](#)
9. Li, J., Li, D., Savarese, S., Hoi, S.: BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In: International Conference on Machine Learning (ICML) (2023) [8](#)
10. Li, J., Li, D., Xiong, C., Hoi, S.: BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In: International Conference on Machine Learning (ICML) (2022) [8](#)
11. Liang, V., Zhang, Y., Kwon, Y., Yeung, S., Zou, J.: Mind the gap: Understanding the modality gap in multi-modal contrastive representation learning. In: Advances in Neural Information Processing Systems (NeurIPS) (2022) [3](#)
12. Lin, T., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., Zitnick, C.: Microsoft coco: Common objects in context. In: European Conference on Computer Vision (ECCV) (2014) [7](#)
13. Parcalabescu, L., Cafagna, M., Muradjan, L., Frank, A., Calixto, I., Gatt, A.: Valse: A task-independent benchmark for vision and language concept evaluation. In: Annual Meeting of the Association for Computational Linguistics (ACL) (2022) [3](#)
14. Patel, M., Sherborn, E., Zhang, M., Jain, S.: Tripletclip: Improving compositional reasoning of clip via synthetic vision-language negatives. In: Advances in Neural Information Processing Systems (NeurIPS) (2024) [3](#), [7](#), [8](#)
15. Peleg, A., Herzig, R., Darrell, T., Globerson, A.: Clic: Contrastive language-image clustering. *arXiv preprint arXiv:2312.06953* (2023) [3](#), [7](#)
16. Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., Sutskever, I.: Learning transferable visual models from natural language supervision. In: International Conference on Machine Learning (ICML). pp. 8748–8763 (2021) [1](#), [7](#), [8](#)
17. Schuhmann, C., Beaumont, R., Vencu, R., Gordon, C., Wightman, R., Cherti, M., Coombes, T., Katta, A., Mullis, C., Wortsman, M., Schramowski, P., Kundurthy, S., Crowson, K., Schmidt, L., Kaczmarczyk, R., Jitsev, J.: LAION-5B: An open large-scale dataset for training next generation image-text models. In: Advances in Neural Information Processing Systems (NeurIPS) (2022) [8](#)

18. Thrush, T., Jiang, R., Bartholomew, M., Ross, A., Bhatt, R., Potts, C., Kiela, D.: Winoground: Probing vision and language models for visuo-linguistic compositionality. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 5238–5248 (2022) 2, 3, 7
19. Yang, Z., Gao, J., Wu, H., Shao, J., Zheng, S., Shi, J.: Structure-clip: Towards scene graph knowledge to enhance multi-modal structured representations. In: AAAI Conference on Artificial Intelligence (2024) 3, 8
20. Yuksekgonul, M., Bianchi, F., Kalluri, P., Jurafsky, D., Zou, J.: When and why vision-language models behave like bags-of-words, and what to do about it. In: International Conference on Learning Representations (ICLR) (2023) 3, 7, 8
21. Zhang, Y., Jiang, C., Feng, F., He, X., Chua, T.: Contrasting intra-modal and ranking cross-modal hard negatives to enhance visio-linguistic compositionality. In: Findings of ACL (2023) 8
22. Zhao, B., Wu, B., Tang, S., Wang, R., Wang, Y., Zhu, J.: Vl-checklist: Evaluating pre-trained vision-language models with objects, attributes and relations. arXiv preprint arXiv:2207.00221 (2022) 3